The explicit classifications at defects one to four

Human Compression project: an attributed recovery of L042

We recover the principal classification in Fernandes, Leemans and Mixer, *An extension of the classification of high rank regular polytopes*, arXiv:1509.01032v1, 3 September 2015. The fixed edition has 26 pages; the subsequent journal citation is *Transactions of the American Mathematical Society* 370 (2018), 8833–8857. The explicit diagrams, and not merely their numbers, form part of the result proved here. The classification theorem and its original families remain attributed to those authors.

A marking is an ordered tuple of nonidentity involutions satisfying the string relations and the intersection property. We identify two markings by an isomorphism preserving their ordered distinguished generators, and also identify a marking with its reverse. Its rank is the number of generators. The *defect* of a degree-n, rank-r marking is $k = n - r$.

Our acquired inputs are the complete transitive rank reduction and fracture theorem, the theorem on subdivision and contraction at a pure perfect cut, and the classification by balanced pairs of rooted wings. Their complete written proofs and accepted finite certificates are declared dependencies. The first uses the recovered L047 and L044 rank classifications. The latter two are the new proofs used in the L019 recovery. In particular, no count or completeness assertion from L042 is used as a premise.

Theorem 1. *Let $1 \leq k \leq 4$, $n \geq 2k + 3$, and let a transitive permutation group of degree n have a connected string C-marking of rank $n - k$. Its group is S_n. Up to marked isomorphism and reversal, the marking belongs to exactly one of the families in Table 1. There are respectively $1, 1, 7, 9$ families for $k = 1, 2, 3, 4$. All these families exist for every degree in the stated range.*

For $n \geq 7$ this is also the classification of the given natural permutation markings up to point conjugacy and reversal. For $n = 5, 6$ the only case is $k = 1$ and the marking is the simplex up to marked group isomorphism; in degree six one chooses its ordinary action to obtain the displayed graph. In degrees $n \geq 7$ every maximal omission of a marking in this range is intransitive.

We give precise conventions for the table so that every edge of every source diagram is specified. Put $m = n - k - 1$. A word such as

$$0, 1, \{0, 2\}, 1, 2, \ldots, m$$

denotes a path with the indicated edge labels; a set in a position means parallel edges on that pair of vertices. Consecutive ascending ranges include their two endpoints. Thus the word just displayed has initial entries $0, 1, \{0, 2\}$ followed by the range $1, \ldots, m$. The following five additional graphs have vertices in a top row, numbered from left to right starting at zero, and a short lower row. Every edge not specified below is absent:

- A_m: top word $2, 1, 2, 3, \ldots, m$; two lower vertices joined by label 2; vertical label-0 edges from top vertices $0, 1$ to the corresponding lower vertices.
- G_m: top word $0, 1, \ldots, m$; two lower vertices joined by label 0; two parallel vertical edges of labels $1, 2$ at top vertex 0, and a vertical edge of label 2 at top vertex 1.
- B_m: top word $0, 1, \ldots, m$; three lower vertices with word $0, 1$; vertical label-3 edges at top vertices $0, 1, 2$.

- E_m: top word $0, 1, \ldots, m, m-1$; two lower vertices joined by label 0; vertical label-2 edges at top vertices $0, 1$.
- I_m: the graph E_m with one further vertical edge, of label 1, at top vertex 0.

These definitions have exactly n vertices at the defects assigned in the table. They also resolve the ordinary convention for shortening the dotted paths at the smallest permitted degree. Our documentary adapter parses every arrow and label of the original Figure 1 at degree 31, expands its dotted paths, and checks literal point conjugacies to these definitions. The original native figure has also been inspected directly.

Table 1: The source families, with a pure cut used for subdivision.

k	Source family	Word or graph	Cut j
1	simplex	$0, 1, \ldots, m$	1
2	rank $n-2$	$1, 0, 1, 2, \ldots, m$	2
3	A	A_m	3
3	G	G_m	3
3	K	$1, 0, 1, 2, \ldots, m, m-1$	2
3	L_2	$2, 1, 0, 1, 2, \ldots, m$	3
3	L_1	$0, 1, 0, 1, 2, \ldots, m$	2
3	M_2, $i = 0$	$0, 1, \{0, 2\}, 1, 2, \ldots, m$	3
3	L_3	$\{0, 2\}, 1, 0, 1, 2, \ldots, m$	3
4	B	B_m	4
4	E	E_m	3
4	I_2	I_m	3
4	N_1	$0, 1, 0, 1, 2, \ldots, m, m-1$	2
4	N_2	$2, 1, 0, 1, 2, \ldots, m, m-1$	3
4	N_3	$\{0, 2\}, 1, 0, 1, 2, \ldots, m, m-1$	3
4	O_2, $i = 0$	$0, 1, \{0, 2\}, 1, 2, \ldots, m, m-1$	3
4	P_1	$1, 0, 1, 0, 1, 2, \ldots, m$	2
4	Q_2	$3, 2, 1, 0, 1, 2, 3, \ldots, m$	4

The names refer to Table 2 and Section 4 of the original article. We write I_2 following Proposition 4.15; the corresponding graph in Table 2 and the concluding list is labelled simply (I). The first nine rows of our table occur in the left column of Figure 1, and the last nine in its right column, in the same vertical order.

Lemma 2. *For each row of Table 1, its edge with label j is a pure perfect cut at every $n \geq 2k+3$. Subdividing this edge by a new vertex, replacing its label by the consecutive labels $j, j+1$, and increasing every former label greater than j by one gives the same family in degree $n+1$, up to ordered point conjugacy.*

Proof. The edge lies in the ascending path between the fixed left part and the right end. Its label occurs exactly once. Removing it separates the graph into two connected parts. All edges with smaller labels lie in the left part, and all edges with larger labels in the right part; parallel edges and the short lower row occur wholly on the appropriate side. These statements follow directly from the table. For completeness, the only right parts with a repeated label have terminal word $m-1, m, m-1$. At the smallest permitted degree, each chosen j is strictly less than $m-1$; the inequalities persist as n increases. Every other repeated or vertical label is smaller than j. Thus neither exception crosses the chosen edge.

The omission has exactly the two connected parts as its orbits, because orbits of permutation generators are precisely the connected components of their labelled graph. The smaller and

larger marks act on the respective parts, proving that the cut is perfect. Its generator is the single transposition on that edge, so it is pure. Subdividing adds one entry to the ascending range. The left part is unchanged, the labels of the right part increase by one, and m changes to $m+1$. These are exactly the defining formulas in the table, including the terminal repeated label. Map the old vertices in path order and the lower vertices to their corresponding vertices; the inserted vertex fills the new path position. This is an explicit ordered conjugacy for every n, not an inference from finitely many numerical checks. □

Lemma 3. *At degree $n = 2k+3$, the rows of Table 1 give precisely the complete classes furnished by the balanced-pair classification, for each $k = 1, 2, 3, 4$.*

Proof. We acquire the complete rooted-wing census and its independent review, rather than repeating that enumeration. A rank-p wing is an actual transitive fractured C-marking on $2p$ points, rooted at a point fixed by every mark except the first, and satisfying the full kernels for adjoining a pure leaf at that root. The accepted census has $1, 1, 6, 3$ rooted point-conjugacy classes for $p = 1, 2, 3, 4$. Write them $W_{p,s}$, with s one plus their zero-based index in the retained literal census. Point conjugacy, not abstract automorphism of a six-point group, is the convention.

The balanced-pair theorem gives one class for every unordered pair of these wings with ranks summing to $k+1$. It provides literal full markings obtained by putting the two roots on opposite sides of a new middle vertex. The certificate in this application records an explicit point bijection, and whether the target marking is reversed, between each smallest source diagram and the following pair:

$k = 1$	simplex : $(W_{1,1}, W_{1,1})$
$k = 2$	rank $n-2$: $(W_{1,1}, W_{2,1})$
$k = 3$	$A : (W_{1,1}, W_{3,5})$, $G : (W_{1,1}, W_{3,6})$, $K : (W_{2,1}, W_{2,1})$, $L_2 : (W_{1,1}, W_{3,2})$, $L_1 : (W_{1,1}, W_{3,1})$, $M_2(0) : (W_{1,1}, W_{3,4})$, $L_3 : (W_{1,1}, W_{3,3})$
$k = 4$	$B : (W_{1,1}, W_{4,3})$, $E : (W_{2,1}, W_{3,5})$, $I_2 : (W_{2,1}, W_{3,6})$, $N_1 : (W_{2,1}, W_{3,1})$, $N_2 : (W_{2,1}, W_{3,2})$, $N_3 : (W_{2,1}, W_{3,3})$, $O_2(0) : (W_{2,1}, W_{3,4})$, $P_1 : (W_{1,1}, W_{4,1})$, $Q_2 : (W_{1,1}, W_{4,2})$.

Each bijection is verified against every point image of every mark. Its discovery and verification require only the literal permutations: assign the image of one point, propagate under the ordered generators, and reject conflicts. Since the action is transitive, every possible conjugacy is obtained by trying all images of that point. Reversing the target covers duality. Each row matches exactly one acquired class, and for each k every class occurs exactly once. This proves the statement, including existence, IP and distinctness, by the already proved properties of those complete acquired classes. □

Proof of Theorem 1. The threshold implies

$$2r = 2(n-k) \geq n+3 > n+2.$$

The acquired transitive fracture theorem therefore makes the group the natural S_n and all maximal omissions intransitive for $n \geq 7$. It also supplies the simplex conclusion and the stated choice of action for degrees five and six. Its proof contains the exact small exceptions to the rank bounds; no generic half-degree estimate is being substituted for them here.

At $n = 2k+3$, Lemma 3 gives all classes and their stated numbers. The acquired stability bijection identifies the classes in each subsequent degree with those in the preceding degree by subdivision at an internal pure perfect cut. It also proves that using any internal pure cut gives the same class, including under reversal. Lemma 2 identifies this operation with growth inside

each displayed family. Induction on n now proves existence, exhaustion and pairwise distinctness of the table. For $n \geq 7$, every automorphism of S_n is inner, so an ordered marked isomorphism is point conjugacy. The explicit degree-six qualification completes the proof in the only exceptional action degree. □

Source contract and qualifications. Theorem 1 recovers Theorem 1.1, Figure 1, the two principal counts and the symmetric-group conclusion of the abstract. It also recovers the intransitive-omission conclusion of Corollary 5.2 for $n \geq 7$, with the necessary ordinary-action choice when $n = 6$. In degree six an outer image of the simplex is a different natural permutation tuple with transitive endpoint omissions; the acquired small census already exhibits it. Thus a literal statement about every degree-six action would be false, although the classification up to marked group isomorphism is unchanged. The original theorem's printed $1 \geq i \geq 4$ is read as $1 \leq i \leq 4$, consistently with the abstract and subsequent propositions.

The other cells of the introductory Table 1 are historical ancillary census reports and are not asserted as additional principal results of this recovery. The 38 candidate families and their elimination form the original proof route; we replace that route by the complete balanced-pair theorem and the explicit identifications above. We do not claim an independent recovery of every intermediate lemma about arbitrary candidate graphs. The introductory higher-defect conjectural direction is covered by the attributed L019 stability result and is not presented as a newly solved open conjecture here.

The new contribution in this application is the exact conversion from rooted wings and pure subdivision to all eighteen source diagrams. The generic theory, finite census and their original attributions are reused through their accepted editions. No compression figure, new Lean result or separate credit for a further original is added.